

Amedeo Carraro

AI Engineer

Vicenza, Italy | Italian (EU) citizen | amedeo.carraro01@gmail.com
amedeocarraro.com | linkedin.com/in/amedeo-carraro | github.com/AmeCarraro

AI Engineer building LLM systems for real operations: retrieval-augmented generation (RAG), tool-calling agents and real-time voice agents, on self-hosted and cloud models. At Terya (TeamSystem Group) I work on the AI side of five such systems, from retrieval and prompts to evaluation and deployment. MSc in Computer Engineering (Artificial Intelligence and Robotics), with a background in deep learning for time series and computer vision.

PROFESSIONAL EXPERIENCE

AI Engineer, Terya (TeamSystem Group), Bassano del Grappa, Italy Nov 2025 – Present

- Service-desk RAG assistant (FastAPI, PostgreSQL/pgvector): structure-aware chunking over several hundred manuals, hybrid dense + full-text search with reciprocal rank fusion and cross-encoder reranking; step-by-step answers cite their sources and include the manuals' original images; two-level knowledge base and an end-to-end test harness that replays real support cases.
- Voice agent for support calls (Python asyncio, OpenAI Realtime API): speech-to-speech with function calling and barge-in; identifies the caller's store despite transcription errors and opens the help-desk ticket automatically, with noise suppression on the audio.
- Agentic RAG platform over the internal codebase (.NET, PostgreSQL/pgvector, K3s): one tool-calling loop behind several chat modes and an MCP server used from coding assistants; hybrid search with reranking and map-reduce analysis of large files.
- ERP orchestrator (FastAPI, SSE, function calling): tool catalogue discovered at runtime, JSON Schema-validated arguments with model self-correction, confirmation gates on destructive actions, per-call audit trail.
- Self-hosted inference: vLLM plus embedding and reranking models serve all the systems above from an on-premises GPU server, with OpenAI as fallback and usage tracking for every request.

Research Intern, Deep Learning (MSc thesis), University of Padua, Italy Sep 2024 – Mar 2025

- Built a 2D-CNN and a hybrid CNN-LSTM in PyTorch to classify variable stars from Zwicky Transient Facility (ZTF) light curves, based on a review of recent work on astronomical time series.
- Reached 99.95% accuracy (F1 0.9995) with the 2D-CNN on the full 5-class dataset and 88.7% on 10 classes, with 5-fold cross-validation; the CNN-LSTM reached 90.0% while training about 3 times faster.

Full Stack Developer Intern, ITReview (TeamSystem Group), Padua, Italy Nov 2023 – Dec 2023

- Developed web applications with React, Node.js/Express and REST APIs on PostgreSQL and MongoDB.
- Cut database response times by about 15%, measured with EXPLAIN ANALYZE, through indexing and query refactoring.

Deep Learning & DevOps Intern, AppLibra S.r.l., Thiene, Italy Oct 2021 – Dec 2021

- Built and deployed a real-time computer-vision check with YOLOv5 (PyTorch) and OpenCV that detects bottle-cap defects on a production conveyor belt, from dataset annotation to GPU training and Docker-based inference on Linux.
- Set up a private NuGet server (BaGet in Docker) for shared .NET libraries and developed ERP customisations for Microsoft Dynamics NAV and Business Central.

EDUCATION

MSc in Computer Engineering: Artificial Intelligence and Robotics, University of Padua Sep 2022 – Mar 2025

Erasmus+ semester, MSc courses in Artificial Intelligence, University of Cyprus Jan 2024 – Jun 2024

BSc in Computer Engineering, University of Padua Sep 2016 – Mar 2022

TECHNICAL SKILLS

LLMs and agents: RAG (hybrid search, reciprocal rank fusion, reranking), function calling, Model Context Protocol (MCP), OpenAI API, Anthropic Claude, vLLM, Ollama, LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, LoRA/QLoRA fine-tuning

Machine learning: PyTorch, TensorFlow/Keras, scikit-learn, YOLO, OpenCV, CNN and LSTM for time series, MLflow, ONNX

Backend and data: Python (FastAPI, asyncio), C#/.NET, SQL, PostgreSQL/pgvector, Elasticsearch, ChromaDB, MongoDB, REST APIs, JavaScript (React, Node.js)

Infrastructure: Docker, Kubernetes (K3s), Linux, Git, GitHub, GitLab | **Other languages:** Java, C, C++, MATLAB

CERTIFICATIONS AND LANGUAGES

Oracle: Agentic AI Foundations Associate (2026), AI Vector Search Certified Professional (2025), OCI Generative AI Professional (2025), OCI AI Foundations Associate (2025)

IBM: Introduction to Containers, Kubernetes, and OpenShift (2026), Docker Essentials: A Developer Introduction (2026)

Languages: Italian (native), English (C1), French (A2)